Cálculo Numérico - Aspectos Teóricos e Computacionais

Capítulo 1

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Exercícios

1. Converta os seguintes números decimais para sua forma binária:
   $x = 37$ $\quad$ $y = 2345$ $\quad$ $z = 0.1217$

2. Converta os seguintes números binários para sua forma decimal:
   $x = (101101)_2$ $\quad$ $y = (110101011)_2$
   $z = (0.01101)_2$ $\quad$ $w = (0.111111101)_2$

Resolução

1. (a)

$$\begin{aligned}
&37\\
&N_0 = 37 = 2 \cdot 18 + 1 \rightarrow a_0 = 1\\
&N_1 = 18 = 2 \cdot 9 + 0 \rightarrow a_1 = 0\\
&N_2 = 9 = 2 \cdot 4 + 1 \rightarrow a_2 = 1\\
&N_3 = 4 = 2 \cdot 2 + 0 \rightarrow a_3 = 0\\
&N_4 = 2 = 2 \cdot 1 + 0 \rightarrow a_4 = 0\\
&N_5 = 1 = 2 \cdot 0 + 1 \rightarrow a_5 = 1\\
&(37)_10 = (100101)_2
\end{aligned}$$

(b)

$$2345$$
$$N_0 = 2345 = 2 \cdot 1172 + 1 \rightarrow a_0 = 1$$
$$N_1 = 1172 = 2 \cdot 586 + 0 \rightarrow a_1 = 0$$
$$N_2 = 586 = 2 \cdot 293 + 0 \rightarrow a_2 = 0$$
$$N_3 = 293 = 2 \cdot 146 + 1 \rightarrow a_3 = 1$$
$$N_4 = 146 = 2 \cdot 73 + 0 \rightarrow a_4 = 0$$
$$N_5 = 73 = 2 \cdot 36 + 1 \rightarrow a_5 = 1$$
$$N_6 = 36 = 2 \cdot 18 + 0 \rightarrow a_6 = 0$$
$$N_7 = 18 = 2 \cdot 9 + 0 \rightarrow a_7 = 0$$
$$N_8 = 9 = 2 \cdot 4 + 1 \rightarrow a_8 = 1$$
$$N_9 = 4 = 2 \cdot 2 + 0 \rightarrow a_9 = 0$$
$$N_10 = 2 = 2 \cdot 1 + 0 \rightarrow a_10 = 0$$
$$N_11 = 1 = 2 \cdot 0 + 1 \rightarrow a_11 = 1$$
$$(2345)_10 = (100100101001)_2$$

(c)

$$
\begin{aligned}
&0.1217\\
&k=1 \quad 2r_1 = 0.2434 => d_1 = 0 \quad r_2 = 0.2434\\
&k=2 \quad 2r_2 = 0.4868 => d_2 = 0 \quad r_3 = 0.4868\\
&k=3 \quad 2r_3 = 0.9736 => d_3 = 0 \quad r_4 = 0.9736\\
&k=4 \quad 2r_4 = 1.9472 => d_4 = 1 \quad r_5 = 0.9472\\
&k=5 \quad 2r_5 = 1.8944 => d_5 = 1 \quad r_6 = 0.8944\\
&k=6 \quad 2r_6 = 1.7888 => d_6 = 1 \quad r_7 = 0.7888\\
&k=7 \quad 2r_7 = 1.5776 => d_7 = 1 \quad r_8 = 0.5776\\
&k=8 \quad 2r_8 = 1.1552 => d_8 = 1 \quad r_9 = 0.1552\\
&k=9 \quad 2r_9 = 0.3104 => d_9 = 0 \quad r_{10} = 0.3104\\
&k=10 \quad 2r_{10} = 0.6208 => d_{10} = 0 \quad r_{11} = 0.6208\\
&k=11 \quad 2r_{11} = 1.2416 => d_{11} = 1 \quad r_{12} = 0.2416\\
&k=12 \quad 2r_{12} = 0.4832 => d_{12} = 0 \quad r_{13} = 0.4832\\
&k=13 \quad 2r_{13} = 0.9664 => d_{13} = 0 \quad r_{14} = 0.9664\\
&k=14 \quad 2r_{14} = 1.9328 => d_{14} = 1 \quad r_{15} = 0.9328\\
&k=15 \quad 2r_{15} = 1.8656 => d_{15} = 1 \quad r_{16} = 0.8656\\
&k=16 \quad 2r_{16} = 1.7312 => d_{16} = 1 \quad r_{17} = 0.7312\\
&k=17 \quad 2r_{17} = 1.4624 => d_{17} = 1 \quad r_{18} = 0.4624\\
&k=18 \quad 2r_{18} = 0.9248 => d_{18} = 0 \quad r_{19} = 0.9248\\
&k=19 \quad 2r_{19} = 1.8496 => d_{19} = 1 \quad r_{20} = 0.8496\\
&k=20 \quad 2r_{20} = 1.6992 => d_{20} = 1 \quad r_{21} = 0.6992\\
&k=21 \quad 2r_{21} = 1.3984 => d_{21} = 1 \quad r_{22} = 0.3984\\
&k=22 \quad 2r_{22} = 0.7968 => d_{22} = 0 \quad r_{23} = 0.7968\\
&k=23 \quad 2r_{23} = 1.5936 => d_{23} = 1 \quad r_{24} = 0.5936\\
&k=24 \quad 2r_{24} = 1.1872 => d_{24} = 1 \quad r_{25} = 0.1872
\end{aligned}
$$

E assim segue infinitamente, não existindo representação exata.

2. (a)

$$
\begin{aligned}
&(101101)_2\\
&= 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0\\
&= 32 + 0 + 8 + 4 + 0 + 1 = 45
\end{aligned}
$$

(b)

$$
\begin{aligned}
&(110101011)_2 \\
&= 1 \cdot 2^8 + 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 \\
&+ 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \\
&= 256 + 128 + 0 + 32 + 0 + 8 + 0 + 2 + 1 \\
&= 427
\end{aligned}
$$

(c)

$$
\begin{aligned}
&(0.01101)_2 \\
&= 0 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3} + 0 \cdot 2^{-4} + 1 \cdot 2^{-5} \\
&= 0 + 0.25 + 0.125 + 0 + 0.03125 \\
&= 0.40625
\end{aligned}
$$

(d)

$$
\begin{aligned}
&(0.111111101)_2 \\
&= 1 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3} + 1 \cdot 2^{-4} \\
&+ 1 \cdot 2^{-5} + 1 \cdot 2^{-6} + 1 \cdot 2^{-7} + 0 \cdot 2^{-8} + 1 \cdot 2^{-9} \\
&= 0.5 + 0.25 + 0.125 + 0.0625 + 0.03125 + 0.015625 \\
&+ 0.0078125 + 0 + 0.001953125 \\
&= 0.994140625
\end{aligned}
$$